

```
-- =====  
-- Layer ۵: ERP Core Master Data Layer (REVISED) - PART ۱ OF ۵  
-- Main File: ۰۲_Master_Data_Table_Definitions.docx  
-- Database Engine: PostgreSQL ۱۸ Enterprise Configuration  
-- =====
```

```
CREATE EXTENSION IF NOT EXISTS pgcrypto;
```

```
-- ۱. business_partners (ریشه یکپارچه اطلاعات مشتریان، تامین‌کنندگان و پیمانکاران)
```

```
CREATE TABLE business_partners ( business_partner_id UUID PRIMARY KEY DEFAULT  
gen_random_uuid(), -- Multi Tenant Isolation tenant_id UUID NOT NULL, -- Identification code  
VARCHAR(۵۰) NOT NULL, display_name VARCHAR(۲۰۰) NOT NULL, -- Partner Classification  
partner_type SMALLINT NOT NULL DEFAULT ۱, -- ۱: Individual -- ۲: Organization status  
SMALLINT NOT NULL DEFAULT ۱, -- ۱: Active -- ۲: Suspended -- ۳: Blocked -- Optional Parent  
Relationship parent_business_partner_id UUID REFERENCES  
business_partners(business_partner_id) ON DELETE RESTRICT, -- Audit Fields created_at  
TIMESTAMPTZ NOT NULL DEFAULT NOW(), created_by UUID, updated_at TIMESTAMPTZ,  
updated_by UUID, deleted_at TIMESTAMPTZ, deleted_by UUID, row_version BIGINT NOT NULL  
DEFAULT ۱ ); -- Unique Business Partner Code per Tenant CREATE UNIQUE INDEX  
uq_business_partners_tenant_code ON business_partners(tenant_id, code) WHERE deleted_at  
IS NULL; -- Tenant Query Optimization CREATE INDEX idx_business_partners_tenant ON  
business_partners(tenant_id) WHERE deleted_at IS NULL; -- Search Optimization CREATE INDEX  
idx_business_partners_display_name ON business_partners(display_name) WHERE deleted_at  
IS NULL; -- Parent Hierarchy Support CREATE INDEX idx_business_partners_parent ON  
business_partners(parent_business_partner_id) WHERE deleted_at IS NULL;
```

```
-- ۲. persons (پروفایل اطلاعات اشخاص حقیقی متصل به شریک تجاری)
```

```
CREATE TABLE persons (  
person_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

business_partner_id UUID NOT NULL REFERENCES business_partners(business_partner_id) ON DELETE RESTRICT, -- قفل سخت‌گیرانه برای حفظ دیتای مالی

first_name VARCHAR(۱۰۰) NOT NULL,

last_name VARCHAR(۱۰۰) NOT NULL,

national_code VARCHAR(۵۰),

birth_date DATE,

gender SMALLINT,

status SMALLINT NOT NULL DEFAULT ۱,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT ۱

);

CREATE UNIQUE INDEX uq_persons_business_partner ON persons(business_partner_id) WHERE deleted_at IS NULL;

CREATE INDEX idx_persons_national_code ON persons(national_code) WHERE deleted_at IS NULL;

-- ۳. business_partner_organizations (پروفاایل اطلاعات شرکت‌ها و سازمان‌های حقوقی)

CREATE TABLE business_partner_organizations (

business_partner_organization_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

business_partner_id UUID NOT NULL REFERENCES business_partners(business_partner_id) ON DELETE RESTRICT,

legal_name VARCHAR(۲۰۰) NOT NULL,

```

trade_name VARCHAR(200),
registration_number VARCHAR(100),
organization_type SMALLINT,
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE UNIQUE INDEX uq_bp_organizations_business_partner ON
business_partner_organizations(business_partner_id) WHERE deleted_at IS NULL;

```

```

-- =====
-- 4. business_partner_roles
-- Business Partner Role Pattern
-- Supports Customer, Supplier, Contractor, Carrier, Service Provider, etc.
-- =====

```

```

CREATE TABLE business_partner_roles (
    business_partner_role_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    business_partner_id UUID NOT NULL

    REFERENCES business_partners(business_partner_id)

```

ON DELETE RESTRICT,

role_type SMALLINT NOT NULL,

/*

 \ Customer

 ʹ Supplier

 Ʒ Contractor

 ξ Carrier

 ο Service Provider

 ᠒ Manufacturer

 Ⅴ Distributor

 ^ Agent

 ᠑ Employee

 \ . Other

*/

role_code VARCHAR(ο .) NOT NULL,

effective_from TIMESTAMPTZ NOT NULL DEFAULT NOW(),

effective_to TIMESTAMPTZ,

status SMALLINT NOT NULL DEFAULT \ ,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

```
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);  
  
CREATE UNIQUE INDEX uq_business_partner_roles_active  
ON business_partner_roles(  
    business_partner_id,  
    role_type  
)  
WHERE deleted_at IS NULL;  
  
CREATE UNIQUE INDEX uq_business_partner_role_code  
ON business_partner_roles(  
    business_partner_id,  
    role_code  
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_business_partner_roles_partner
ON business_partner_roles(
    business_partner_id
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_business_partner_roles_type
ON business_partner_roles(
    role_type
)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ٥. business_partner_contacts
--   مديریت مخاطبین مستقل Business Partner
-- =====
```

```
CREATE TABLE business_partner_contacts (
    business_partner_contact_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    business_partner_id UUID NOT NULL
    REFERENCES business_partners(business_partner_id)
    ON DELETE RESTRICT,
```

contact_type SMALLINT NOT NULL,

/*

 \ Primary Contact

 ʒ Financial Contact

 Ƴ Technical Contact

 ξ Sales Contact

 ο Support Contact

 Ϛ Legal Contact

*/

first_name VARCHAR(\ . .) NOT NULL,

last_name VARCHAR(\ . .) NOT NULL,

job_title VARCHAR(\ ο .),

email VARCHAR(ʒοο),

mobile VARCHAR(ο .),

phone VARCHAR(ο .),

is_primary BOOLEAN NOT NULL DEFAULT FALSE,

status SMALLINT NOT NULL DEFAULT \,

```
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,

updated_at TIMESTAMPTZ,
updated_by UUID,

deleted_at TIMESTAMPTZ,
deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT \
);
```

```
CREATE INDEX idx_bp_contacts_partner
ON business_partner_contacts(business_partner_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_bp_contacts_email
ON business_partner_contacts(email)
WHERE deleted_at IS NULL;
```

```
CREATE UNIQUE INDEX uq_bp_primary_contact
ON business_partner_contacts(
    business_partner_id,
```



```
        contact_type
    )
    WHERE is_primary = TRUE
    AND deleted_at IS NULL;
```

-- ۵. business_partner_identifications (مدیریت مدارک شناسایی مانند شناسه اقتصادی یا گذرنامه)

```
CREATE TABLE business_partner_identifications (
    bp_identification_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_partner_id UUID NOT NULL REFERENCES business_partners(business_partner_id) ON
    DELETE RESTRICT,
    tenant_id UUID NOT NULL,
    identification_type_code VARCHAR(۵۰) NOT NULL, -- (e.g., 'ECONOMIC_CODE', 'PASSPORT')
    id_number VARCHAR(۱۰۰) NOT NULL,
    issue_date DATE,
    expiry_date DATE,
    status SMALLINT NOT NULL DEFAULT ۱,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    deleted_at TIMESTAMPTZ
);

CREATE UNIQUE INDEX uq_bp_identifications_tenant ON
business_partner_identifications(tenant_id, identification_type_code, id_number) WHERE
deleted_at IS NULL;

CREATE INDEX idx_bp_identifications_partner ON
business_partner_identifications(business_partner_id) WHERE deleted_at IS NULL;
```

-- =====

-- Layer ۵: ERP Core Master Data Layer (REVISED) - PART ۲ OF ۵

-- Main File: ۰۲_Master_Data_Table_Definitions.docx

-- Database Engine: PostgreSQL ۱۸

-- =====

-- ۶. wf_process_definitions (تعریف شابلون فرایندها - خط تولید یا زنجیره خدمات)

```
CREATE TABLE wf_process_definitions (  
    process_definition_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL, -- قفل در بافتار چندمستأجری امن  
    code VARCHAR(۱۰۰) NOT NULL, -- (e.g., 'MANUFACTURING_FLOW_V۱',  
    'CYBER_SERVICE_FLOW')  
    name VARCHAR(۲۰۰) NOT NULL,  
    target_aggregate_type VARCHAR(۱۰۰) NOT NULL, -- (موجودیتی که فرایند رویش اجرا میشود)  
    flow_graph JSONB NOT NULL, -- ذخیره کل مراحل، شرطها و مسیرهای بازگشت به صورت گراف پویا  
    is_active BOOLEAN NOT NULL DEFAULT TRUE,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    row_version BIGINT NOT NULL DEFAULT ۱  
);  
  
CREATE UNIQUE INDEX uq_wf_process_def ON wf_process_definitions(tenant_id, code);
```

-- ۷. wf_process_instances (موتور ردیابی زنده وضعیت هر سند در گردش کار)

```
CREATE TABLE wf_process_instances (  
    process_instance_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL,  
    process_definition_id UUID NOT NULL REFERENCES  
    wf_process_definitions(process_definition_id) ON DELETE RESTRICT,
```

```

target_aggregate_id UUID NOT NULL, -- شناسه فیزیکی فاکتور، سند حسابداری یا سفارش تولید (Logical Reference)

current_state VARCHAR(۱۰۰) NOT NULL, -- وضعیت فعلی در گراف فرایند

owning_tenant_id UUID NOT NULL, -- شرکتی که این مرحله را مالک است (حیاتی برای اتوماسیون بین شرکتی)

status SMALLINT NOT NULL DEFAULT ۱, -- (۱: Running, ۲: Completed, ۳: Terminated/Failed)

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

updated_at TIMESTAMPTZ

);

CREATE INDEX idx_wf_instances_target ON wf_process_instances(target_aggregate_id);

CREATE INDEX idx_wf_instances_owner ON wf_process_instances(owning_tenant_id,
current_state);

-- ۸. wf_tasks (کارپوشه پورتال - ارجاع تسک ها به کارمندان داخلی یا پیمانکاران خارجی بخش اول)

CREATE TABLE wf_tasks (

task_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_id UUID NOT NULL,

process_instance_id UUID NOT NULL REFERENCES wf_process_instances(process_instance_id)
ON DELETE RESTRICT,

assigned_type SMALLINT NOT NULL, -- (۱: Internal Role, ۲: External Business Partner / Contractor)

assigned_to_id UUID NOT NULL, -- business_partner_id لایه ۴ و اگر ۲ باشد role_id اگر ۱ باشد شناسه -- !بخش اول است

task_name VARCHAR(۲۰۰) NOT NULL,

status SMALLINT NOT NULL DEFAULT ۱, -- (۱: Pending, ۲: Approved, ۳: Rejected)

context_snapshots JSONB, -- سنگین JOIN ذخیره اطلاعات خلاصه سند برای نمایش سریع در فرانت اند بدون

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

actioned_at TIMESTAMPTZ,

```

actioned_by UUID

);

CREATE INDEX idx_wf_tasks_assignment ON wf_tasks(tenant_id, assigned_type, assigned_to_id) WHERE status = \;

-- ۹. entity_addresses (جدول آدرس‌های چندشکلی چندمستأجری برای شرکا، شعب یا شرکت‌ها)

CREATE TABLE entity_addresses (

entity_address_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_id UUID NOT NULL,

entity_type VARCHAR(۱۰۰) NOT NULL, -- (e.g., 'BUSINESS_PARTNER', 'BRANCH', 'COMPANY')

entity_id UUID NOT NULL, -- شناسه پویای لایه‌ای

address_type_id UUID NOT NULL, -- متصل به مقادیر پایه بخش چهارم

country_id UUID NOT NULL, -- ارجاع منطقی گلوبال به بخش چهارم

province_name VARCHAR(۱۵۰),

city_name VARCHAR(۱۵۰),

postal_code VARCHAR(۵۰),

address_text TEXT NOT NULL,

is_primary BOOLEAN NOT NULL DEFAULT FALSE,

status SMALLINT NOT NULL DEFAULT \,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT \

);

CREATE INDEX idx_entity_addresses_polymorphic ON entity_addresses(entity_id, entity_type)
WHERE deleted_at IS NULL;

CREATE INDEX idx_entity_addresses_tenant ON entity_addresses(tenant_id) WHERE deleted_at
IS NULL;

-- ۱۰. entity_contact_points (جدول چندشکلی اطلاعات تماس - تلفن، ایمیل، وبسایت)

CREATE TABLE entity_contact_points (
entity_contact_point_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
entity_type VARCHAR(۱۰۰) NOT NULL,
entity_id UUID NOT NULL,
contact_point_type SMALLINT NOT NULL, -- (۱: Phone, ۲: Mobile, ۳: Email, ۴: Website)
contact_value VARCHAR(۲۵۵) NOT NULL,
is_primary BOOLEAN NOT NULL DEFAULT FALSE,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE INDEX idx_entity_contact_polymorphic ON entity_contact_points(entity_id,
entity_type) WHERE deleted_at IS NULL;

```
-- =====  
-- Layer ۵: ERP Core Master Data Layer (REVISED) - PART ۳ OF ۵  
-- Main File: ۰۲_Master_Data_Table_Definitions.docx  
-- Database Engine: PostgreSQL ۱۸  
-- =====
```

-- ۱۱. fiscal_years (مدیریت سال‌های مالی هر شرکت به صورت ایزوله چندمستأجری)

```
CREATE TABLE fiscal_years (  
    fiscal_year_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL, -- متصل به بافتار مالی-تنت امن پلتفرم  
    company_id UUID NOT NULL, -- فیزیکی بر اساس اصول مرزبندی FK بدون) ارجاع منطقی به لایه ۴  
    year_code VARCHAR(۵۰) NOT NULL, -- (e.g., "۱۴۰۵", "۲۰۲۶")  
    name VARCHAR(۲۰۰) NOT NULL,  
    start_date DATE NOT NULL,  
    end_date DATE NOT NULL,  
    is_closed BOOLEAN NOT NULL DEFAULT FALSE,  
    status SMALLINT NOT NULL DEFAULT ۱,  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    created_by UUID,  
    updated_at TIMESTAMPTZ,  
    updated_by UUID,  
    deleted_at TIMESTAMPTZ,  
    deleted_by UUID,  
    row_version BIGINT NOT NULL DEFAULT ۱  
);
```

```
CREATE UNIQUE INDEX uq_fiscal_years_tenant_code ON fiscal_years(tenant_id, company_id,
year_code) WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_fiscal_years_company ON fiscal_years(company_id) WHERE deleted_at IS
NULL;
```

-- ۱۲. fiscal_periods (ماه یا دوره‌های مالی عملیاتی درون سال مالی)

```
CREATE TABLE fiscal_periods (
```

```
    fiscal_period_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    tenant_id UUID NOT NULL,
```

```
    fiscal_year_id UUID NOT NULL REFERENCES fiscal_years(fiscal_year_id) ON DELETE RESTRICT, --  
    قفل از حذف آبشاری زنجیره‌ای مالی
```

```
    period_number SMALLINT NOT NULL, -- (۱-۱۲)
```

```
    name VARCHAR(۱۰۰) NOT NULL,
```

```
    start_date DATE NOT NULL,
```

```
    end_date DATE NOT NULL,
```

```
    is_closed BOOLEAN NOT NULL DEFAULT FALSE,
```

```
    status SMALLINT NOT NULL DEFAULT ۱,
```

```
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
```

```
    created_by UUID,
```

```
    updated_at TIMESTAMPTZ,
```

```
    updated_by UUID,
```

```
    deleted_at TIMESTAMPTZ,
```

```
    deleted_by UUID,
```

```
    row_version BIGINT NOT NULL DEFAULT ۱
```

```
);
```

```
CREATE UNIQUE INDEX uq_fiscal_periods_sequence ON fiscal_periods(fiscal_year_id,
period_number) WHERE deleted_at IS NULL;
```

-- ۱۳. shipment_methods (روش‌های حمل و تحویل کالا کاتالوگ لوکاپ)

```
CREATE TABLE shipment_methods (  
  shipment_method_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  code VARCHAR(۵۰) NOT NULL, -- (e.g., "AIR", "SEA", "EXPRESS_POST")  
  name VARCHAR(۱۵۰) NOT NULL,  
  description VARCHAR(۵۰۰),  
  status SMALLINT NOT NULL DEFAULT ۱,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  deleted_at TIMESTAMPTZ  
);  
  
CREATE UNIQUE INDEX uq_shipment_methods_tenant_code ON shipment_methods(tenant_id,  
code) WHERE deleted_at IS NULL;
```

-- ۱۴. payment_terms (ترم‌ها و شرایط پرداخت فاکتورها - سقف اعتباری و نقدی)

-- برای مبالغ و نرخ جریمه‌ها جهت تضمین صحت محاسبات (۲۰,۴) NUMERIC(۲۰,۴) اصلاح محاسباتی: ارتقا به

```
CREATE TABLE payment_terms (  
  payment_term_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  code VARCHAR(۵۰) NOT NULL, -- (e.g., "NET_۳۰", "۲_۱۰_NET_۳۰")  
  name VARCHAR(۲۰۰) NOT NULL,  
  description VARCHAR(۵۰۰),  
  net_days INT NOT NULL DEFAULT ۰,  
  discount_days INT NOT NULL DEFAULT ۰,  
  discount_percentage NUMERIC(۲۰,۴) NOT NULL DEFAULT ۰,
```



```

penalty_percentage_per_month NUMERIC(۲۰,۴) NOT NULL DEFAULT ۰,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE UNIQUE INDEX uq_payment_terms_tenant_code ON payment_terms(tenant_id, code)
WHERE deleted_at IS NULL;

```

```

-- =====
-- Layer ۵: ERP Core Master Data Layer (REVISED) - PART ۴ OF ۵
-- Main File: ۰۲_Master_Data_Table_Definitions.docx
-- Database Engine: PostgreSQL ۱۸
-- =====

```

```

-- ۱۵. units_of_measure (واحد‌های سنجش انبار و کالا)
-- اضافه شد و ایندکس ترکیبی شد تا تداخل گلوبال رخ ندهد tenant_id اصلاح امنیت ساس: ستون
CREATE TABLE units_of_measure (
uom_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
code VARCHAR(۵۰) NOT NULL, -- (e.g., "KG", "BOX", "METER")
name VARCHAR(۱۰۰) NOT NULL,

```

```

decimal_places SMALLINT NOT NULL DEFAULT ۰,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,
deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT ۱
);

CREATE UNIQUE INDEX uq_uom_tenant_code ON units_of_measure(tenant_id, code) WHERE
deleted_at IS NULL;

CREATE INDEX idx_uom_tenant ON units_of_measure(tenant_id) WHERE deleted_at IS NULL;

```

-- ۱۶. countries (tenant_id کاتالوگ ثابت و عمومی پلتفرم - بدون نیاز به)

```

CREATE TABLE countries (
country_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
iso_code VARCHAR(۱۰) NOT NULL, -- (e.g., "IR", "DE", "US")
name VARCHAR(۱۵۰) NOT NULL,
phone_code VARCHAR(۲۰),
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
deleted_at TIMESTAMPTZ
);

CREATE UNIQUE INDEX uq_countries_iso ON countries(iso_code) WHERE deleted_at IS NULL;

```

-- =====

-- ٢٠. currencies

-- Currency Master Data

-- =====

CREATE TABLE currencies (

currency_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

code VARCHAR(١٠) NOT NULL,

-- مثال: USD, EUR, IRR

name VARCHAR(١٠٠) NOT NULL,

symbol VARCHAR(١٠),

decimal_places SMALLINT NOT NULL DEFAULT ٢,

is_base_currency BOOLEAN NOT NULL DEFAULT FALSE,

status SMALLINT NOT NULL DEFAULT ١,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

```
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_currencies_code  
ON currencies(code)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_currencies_status  
ON currencies(status)  
WHERE deleted_at IS NULL;
```

```
-- =====  
-- 2\ . exchange_rates  
-- Multi Currency Exchange Rate Master Data  
-- =====
```

```
CREATE TABLE exchange_rates (  
    exchange_rate_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    from_currency_id UUID NOT NULL
```

REFERENCES currencies(currency_id)

ON DELETE RESTRICT,

to_currency_id UUID NOT NULL

REFERENCES currencies(currency_id)

ON DELETE RESTRICT,

rate NUMERIC(10,4) NOT NULL,

effective_from TIMESTAMPTZ NOT NULL DEFAULT NOW(),

effective_to TIMESTAMPTZ,

rate_type SMALLINT NOT NULL DEFAULT 1,

/*

1 Official

2 Market

3 Custom

*/

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

```
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);  
  
CREATE INDEX idx_exchange_rates_from_currency  
ON exchange_rates(from_currency_id)  
WHERE deleted_at IS NULL;  
  
CREATE INDEX idx_exchange_rates_to_currency  
ON exchange_rates(to_currency_id)  
WHERE deleted_at IS NULL;  
  
CREATE UNIQUE INDEX uq_exchange_rate_period  
ON exchange_rates(  
    from_currency_id,  
    to_currency_id,  
    effective_from  
)  
WHERE deleted_at IS NULL;
```

```
-- =====  
-- ୧୧. chart_of_accounts  
-- Financial Account Master Data  
-- =====
```

```
CREATE TABLE chart_of_accounts (  
    account_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    tenant_id UUID NOT NULL,  
  
    company_id UUID NOT NULL,  
  
    parent_account_id UUID  
        REFERENCES chart_of_accounts(account_id)  
        ON DELETE RESTRICT,  
  
    account_code VARCHAR(୧୦) NOT NULL,  
  
    account_name VARCHAR(୧୦୦) NOT NULL,  
  
    account_type SMALLINT NOT NULL,  
  
    /*  
        ୧ Asset  
        ୨ Liability  
        ୩ Equity
```

ξ Revenue

ο Expense

*/

account_level SMALLINT NOT NULL DEFAULT \,

is_postable BOOLEAN NOT NULL DEFAULT TRUE,

-- آیا سند مالی مستقیم روی این حساب ثبت می‌شود --

normal_balance SMALLINT NOT NULL DEFAULT \,

/*

\ Debit

۲ Credit

*/

status SMALLINT NOT NULL DEFAULT \,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,


```
    row_version BIGINT NOT NULL DEFAULT 1
);
```

```
CREATE UNIQUE INDEX uq_chart_accounts_code
ON chart_of_accounts(
    tenant_id,
    company_id,
    account_code
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_chart_accounts_parent
ON chart_of_accounts(parent_account_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_chart_accounts_company
ON chart_of_accounts(company_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ୨୩. banks
-- Financial Master Data - Bank Directory
```

-- =====

```
CREATE TABLE banks (  
    bank_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    code VARCHAR(3) NOT NULL,  
  
    name VARCHAR(255) NOT NULL,  
  
    country_id UUID,  
  
    swift_code VARCHAR(8),  
  
    status SMALLINT NOT NULL DEFAULT 1,  
  
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
    created_by UUID,  
  
    updated_at TIMESTAMPTZ,  
    updated_by UUID,  
  
    deleted_at TIMESTAMPTZ,  
    deleted_by UUID,  
  
    row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_banks_code
ON banks(code)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_banks_country
ON banks(country_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_banks_status
ON banks(status)
WHERE deleted_at IS NULL;
```

```
-- =====
```

```
-- ႃႃ. items
```

```
-- Inventory Master Data - Core Item Definition
```

```
-- =====
```

```
CREATE TABLE items (
    item_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,
```

item_category_id UUID,

uom_id UUID NOT NULL,

code VARCHAR(6) NOT NULL,

name VARCHAR(255) NOT NULL,

item_type SMALLINT NOT NULL,

/*

 1 Stock Item

 2 Service

 3 Non Stock Item

 4 Consumable

 5 Asset

*/

description TEXT,

is_inventory_item BOOLEAN NOT NULL DEFAULT TRUE,

is_purchase_item BOOLEAN NOT NULL DEFAULT TRUE,

is_sales_item BOOLEAN NOT NULL DEFAULT FALSE,

track_serial_number BOOLEAN NOT NULL DEFAULT FALSE,

```
track_batch_number BOOLEAN NOT NULL DEFAULT FALSE,  
  
status SMALLINT NOT NULL DEFAULT \,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT \  
);
```

```
CREATE UNIQUE INDEX uq_items_tenant_code  
ON items(  
    tenant_id,  
    code  
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_items_category
ON items(item_category_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_items_uom
ON items(uom_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_items_status
ON items(status)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ㄱ. item_categories
-- Inventory Master Data - Item Classification
-- =====
```

```
CREATE TABLE item_categories (
    item_category_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    parent_category_id UUID
```

REFERENCES item_categories(item_category_id)

ON DELETE RESTRICT,

code VARCHAR(50) NOT NULL,

name VARCHAR(255) NOT NULL,

description VARCHAR(500),

category_type SMALLINT NOT NULL DEFAULT 1,

/*

1 Stock

2 Service

3 Asset

4 Consumable

*/

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

```
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_item_categories_tenant_code  
ON item_categories(  
    tenant_id,  
    code  
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_item_categories_parent  
ON item_categories(parent_category_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_item_categories_tenant  
ON item_categories(tenant_id)  
WHERE deleted_at IS NULL;
```

```
-- =====  
-- ㄱ. products  
-- Inventory Master Data - Sales Product Definition
```


-- =====

```
CREATE TABLE products (  
    product_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    tenant_id UUID NOT NULL,  
  
    item_id UUID NOT NULL  
        REFERENCES items(item_id)  
        ON DELETE RESTRICT,  
  
    product_code VARCHAR(50) NOT NULL,  
  
    product_name VARCHAR(255) NOT NULL,  
  
    product_type SMALLINT NOT NULL DEFAULT 1,  
    /*  
        1 Finished Product  
        2 Service Product  
        3 Subscription Product  
        4 Bundle Product  
    */  
  
    sales_description TEXT,  
  
    purchase_description TEXT,
```

```
default_sales_price NUMERIC(ȳ·,ȳ),

default_purchase_price NUMERIC(ȳ·,ȳ),

tax_category_id UUID,

status SMALLINT NOT NULL DEFAULT \,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT \

);
```

```
CREATE UNIQUE INDEX uq_products_tenant_code

ON products(

    tenant_id,

    product_code
```

```
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_products_item  
ON products(item_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_products_tax_category  
ON products(tax_category_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_products_status  
ON products(status)  
WHERE deleted_at IS NULL;
```

```
-- =====
```

```
-- Ⅳ. warehouses
```

```
-- Inventory Master Data - Warehouse Definition
```

```
-- =====
```

```
CREATE TABLE warehouses (  
    warehouse_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

tenant_id UUID NOT NULL,

company_id UUID NOT NULL,

code VARCHAR(8) NOT NULL,

name VARCHAR(255) NOT NULL,

warehouse_type SMALLINT NOT NULL DEFAULT 1,

/*

1 Main Warehouse

2 Branch Warehouse

3 Transit Warehouse

4 Virtual Warehouse

*/

address_id UUID,

manager_user_id UUID,

allow_negative_stock BOOLEAN NOT NULL DEFAULT FALSE,

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

```
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);  
  
CREATE UNIQUE INDEX uq_warehouses_tenant_code  
ON warehouses(  
    tenant_id,  
    code  
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_warehouses_company  
ON warehouses(company_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_warehouses_status
```

ON warehouses(status)

WHERE deleted_at IS NULL;

-- =====

-- ㄱ. warehouse_locations

-- Inventory Master Data - Bin / Storage Location Definition

-- =====

CREATE TABLE warehouse_locations (

warehouse_location_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_id UUID NOT NULL,

warehouse_id UUID NOT NULL

REFERENCES warehouses(warehouse_id)

ON DELETE RESTRICT,

parent_location_id UUID

REFERENCES warehouse_locations(warehouse_location_id)

ON DELETE RESTRICT,

code VARCHAR(5) NOT NULL,

name VARCHAR(25) NOT NULL,

```
location_type SMALLINT NOT NULL DEFAULT \,  
/*  
    \ Zone  
    ʹ Aisle  
    ʻ Rack  
    ξ Shelf  
    ° Bin  
*/  
  
is_storage_location BOOLEAN NOT NULL DEFAULT TRUE,  
  
capacity_quantity NUMERIC(ʹ·,ξ),  
  
status SMALLINT NOT NULL DEFAULT \,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT \  
);
```

```
CREATE UNIQUE INDEX uq_warehouse_locations_code
ON warehouse_locations(
    warehouse_id,
    code
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_warehouse_locations_parent
ON warehouse_locations(parent_location_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_warehouse_locations_warehouse
ON warehouse_locations(warehouse_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ㄲ. tax_definitions
-- Financial Master Data - Tax Rules Definition
-- =====
```

```
CREATE TABLE tax_definitions (
```


tax_definition_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_id UUID NOT NULL,

tax_category_id UUID

REFERENCES tax_categories(tax_category_id)

ON DELETE RESTRICT,

code VARCHAR(10) NOT NULL,

name VARCHAR(255) NOT NULL,

tax_type SMALLINT NOT NULL,

/*

 \ VAT

 \ Sales Tax

 \ Purchase Tax

 \ Withholding Tax

 \ Other

*/

calculation_type SMALLINT NOT NULL DEFAULT \,

/*

 \ Percentage

 \ Fixed Amount

*/

```
tax_rate NUMERIC(۲۰,۴) NOT NULL DEFAULT ۰,  
  
fixed_amount NUMERIC(۲۰,۴),  
  
effective_from TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  
effective_to TIMESTAMPTZ,  
  
is_inclusive BOOLEAN NOT NULL DEFAULT FALSE,  
    -- آیا مالیات داخل مبلغ نهایی محاسبه شده است --  
  
status SMALLINT NOT NULL DEFAULT ۱,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT ۱  
);
```

```
CREATE UNIQUE INDEX uq_tax_definitions_tenant_code
ON tax_definitions(
    tenant_id,
    code
)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tax_definitions_category
ON tax_definitions(tax_category_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tax_definitions_status
ON tax_definitions(status)
WHERE deleted_at IS NULL;
```

-- ۱۷. address_types (انواع متادیتای آدرس - متصل به جدول چندشکلی آدرس‌ها)

```
CREATE TABLE address_types (
    address_type_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL,
    code VARCHAR(۵۰) NOT NULL, -- (e.g., "OFFICE", "FACTORY", "WAREHOUSE_DELIVERY")
```

```

name VARCHAR(۱۰۰) NOT NULL,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
deleted_at TIMESTAMPTZ
);

CREATE UNIQUE INDEX uq_address_types_tenant_code ON address_types(tenant_id, code)
WHERE deleted_at IS NULL;

CREATE INDEX idx_address_types_tenant ON address_types(tenant_id) WHERE deleted_at IS
NULL;

```

-- ۱۸. bank_accounts (مدیریت اطلاعات و شماره حساب‌های بانکی شرکت‌ها یا شرکای تجاری)

```

CREATE TABLE bank_accounts (
bank_account_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
entity_type VARCHAR(۱۰۰) NOT NULL, -- (e.g., 'COMPANY', 'BUSINESS_PARTNER')
entity_id UUID NOT NULL, -- Logical Reference پویا
bank_name VARCHAR(۱۵۰) NOT NULL,
branch_name VARCHAR(۱۵۰),
account_number VARCHAR(۱۰۰) NOT NULL,
card_number VARCHAR(۵۰),
iban VARCHAR(۱۰۰), -- شماره شبا
is_primary BOOLEAN NOT NULL DEFAULT FALSE,
status SMALLINT NOT NULL DEFAULT ۱,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,
updated_at TIMESTAMPTZ,
updated_by UUID,

```

```

deleted_at TIMESTAMPTZ,
deleted_by UUID,
row_version BIGINT NOT NULL DEFAULT 1
);

CREATE INDEX idx_bank_accounts_polymorphic ON bank_accounts(entity_id, entity_type)
WHERE deleted_at IS NULL;

CREATE INDEX idx_bank_accounts_tenant ON bank_accounts(tenant_id) WHERE deleted_at IS
NULL;

```

-- ۱۹. tax_categories (دسته‌بندی‌های کلان قوانین مالیاتی و عوارض کالا)

```

CREATE TABLE tax_categories (
tax_category_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
code VARCHAR(۵۰) NOT NULL, -- (e.g., "VAT_۹", "VAT_۱۰", "EXEMPT")
name VARCHAR(۱۵۰) NOT NULL,
description VARCHAR(۵۰۰),
status SMALLINT NOT NULL DEFAULT 1,
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
deleted_at TIMESTAMPTZ
);

CREATE UNIQUE INDEX uq_tax_categories_tenant_code ON tax_categories(tenant_id, code)
WHERE deleted_at IS NULL;

```

```
-- =====  
-- document_sequences  
-- ERP Document Numbering Engine  
-- Centralized Dynamic Sequence Management  
-- =====
```

```
CREATE TABLE document_sequences (
```

```
    sequence_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    tenant_id UUID NOT NULL,
```

```
    company_id UUID,
```

```
    module_code VARCHAR(5) NOT NULL,
```

```
    /*
```

```
        FIN
```

```
        INV
```

```
        PUR
```

```
        SAL
```

```
        HR
```

```
        MFG
```

```
    */
```

```
    document_type VARCHAR(10) NOT NULL,
```

```
    /*
```

SALES_INVOICE

PURCHASE_ORDER

STOCK_RECEIPT

JOURNAL_ENTRY

*/

document_scope SMALLINT NOT NULL DEFAULT \,

/*

\ Tenant

Ƴ Company

ƴ Branch

ξ Module

*/

owner_type VARCHAR(º·),

/*

COMPANY

BRANCH

MODULE

*/

owner_id UUID,

prefix VARCHAR(Ƴ·),

suffix VARCHAR(Ƴ·),

```
padding_length INT NOT NULL DEFAULT ٦,  
  
current_value BIGINT NOT NULL DEFAULT ٠,  
  
reset_period SMALLINT NOT NULL DEFAULT ١,  
  
/*  
    ١ Never  
    ٢ Yearly  
    ٣ Monthly  
*/  
  
last_reset_at TIMESTAMPTZ,  
  
is_active BOOLEAN NOT NULL DEFAULT TRUE,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
  
updated_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT ١  
);
```



```
CREATE UNIQUE INDEX uq_document_sequences_scope
ON document_sequences(
    tenant_id,
    module_code,
    document_type,
    document_scope,
    COALESCE(company_id, '.....'::UUID)
)
WHERE is_active = TRUE;
```

```
CREATE INDEX idx_document_sequences_module
ON document_sequences(
    tenant_id,
    module_code
)
WHERE is_active = TRUE;
```

```
CREATE INDEX idx_document_sequences_owner
ON document_sequences(owner_id)
WHERE is_active = TRUE;
```

```
-- =====  
-- attachments  
-- Core File Management Service  
-- Shared Attachment Metadata Across ERP Modules  
-- =====
```

```
CREATE TABLE attachments (
```

```
    attachment_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
    tenant_id UUID NOT NULL,
```

```
    entity_type VARCHAR(100) NOT NULL,
```

```
-- نام موجودیت مقصد
```

```
-- مثال: INVOICE, PURCHASE_ORDER, EMPLOYEE
```

```
    entity_id UUID NOT NULL,
```

```
-- شناسه رکورد مقصد
```

```
    file_name VARCHAR(255) NOT NULL,
```

```
    original_file_name VARCHAR(255),
```

```
    file_extension VARCHAR(20),
```

```
    mime_type VARCHAR(100) NOT NULL,
```

file_size_bytes BIGINT NOT NULL,

storage_provider VARCHAR(50) NOT NULL DEFAULT 'LOCAL',

/*

LOCAL

S3

AZURE

MINIO

*/

storage_path VARCHAR(255) NOT NULL,

checksum VARCHAR(64),

description VARCHAR(255),

uploaded_by UUID,

uploaded_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

status SMALLINT NOT NULL DEFAULT 1,

/*

1 Active

2 Archived

3 Deleted

*/

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT 1

);

CREATE INDEX idx_attachments_entity

ON attachments(

tenant_id,

entity_type,

entity_id

)

WHERE deleted_at IS NULL;

```
CREATE INDEX idx_attachments_checksum
ON attachments(checksum)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_attachments_status
ON attachments(status)
WHERE deleted_at IS NULL;
```

```
-- =====
-- tags
-- ERP Dynamic Tagging Master Data
-- Tenant Scoped Tag Management
-- =====
```

```
CREATE TABLE tags (

    tag_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    scope_type VARCHAR(50) NOT NULL,

    /*

    SYSTEM

    MODULE
```

ENTITY

*/

module_code VARCHAR(۵۰),

/*

INV

PUR

SAL

FIN

HR

*/

entity_type VARCHAR(۱۰۰),

/*

برای موجودیت خاص Tag محدودسازی

مثال:

INVOICE

ITEM

CUSTOMER

*/

tag_name VARCHAR(۱۰۰) NOT NULL,

description VARCHAR(۵۰۰),

color_code VARCHAR(۲۰),

```
status SMALLINT NOT NULL DEFAULT \,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT \  
);
```

```
CREATE UNIQUE INDEX uq_tags_scope_name  
ON tags(  
    tenant_id,  
    scope_type,  
    COALESCE(module_code, ''),  
    COALESCE(entity_type, ''),  
    tag_name
```

```
)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tags_tenant  
ON tags(tenant_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tags_module  
ON tags(module_code)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tags_entity_type  
ON tags(entity_type)  
WHERE deleted_at IS NULL;
```

```
-- =====  
-- (جدول واسط چندشکلی نگاشت برچسب‌ها به رکوردهای مختلف سیستم) entity_tags ۴.  
-- =====  
  
CREATE TABLE entity_tags (
```



```

entity_tag_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
tenant_id UUID NOT NULL,
tag_id UUID NOT NULL REFERENCES tags(tag_id) ON DELETE RESTRICT,
target_entity_type VARCHAR(۱۰۰) NOT NULL, -- نام جدول هدف ('pur_orders')
target_entity_id UUID NOT NULL, -- شناسه ردیف هدف
assigned_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
assigned_by UUID
);

```

```

CREATE UNIQUE INDEX uq_entity_tags_mapping ON entity_tags(tenant_id, tag_id,
target_entity_type, target_entity_id);

```

```

CREATE INDEX idx_entity_tags_lookup ON entity_tags(tenant_id, target_entity_type,
target_entity_id);

```

```

-- =====

```

```

-- ۵. employees (جدول اطلاعات و شناسنامه پرسنل سازمان)

```

```

-- =====

```

```

CREATE TABLE employees (
    employee_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    tenant_id UUID NOT NULL,
    user_id UUID, -- ارجاع منطقی اختیاری به جدول کاربران سیستم هویت
    employee_code VARCHAR(۵۰) NOT NULL,
    first_name VARCHAR(۱۰۰) NOT NULL,
    last_name VARCHAR(۱۰۰) NOT NULL,
    national_code VARCHAR(۱۰) NOT NULL,
    hire_date DATE NOT NULL,
    termination_date DATE,

```

```
status SMALLINT NOT NULL DEFAULT ۱, -- ۱: Active, ۲: Suspended, ۳: Terminated
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
updated_at TIMESTAMPTZ,
deleted_at TIMESTAMPTZ,
row_version BIGINT NOT NULL DEFAULT ۱
);
```

```
CREATE UNIQUE INDEX uq_employees_code ON employees(tenant_id, employee_code) WHERE
deleted_at IS NULL;
```

```
CREATE UNIQUE INDEX uq_employees_national ON employees(tenant_id, national_code)
WHERE deleted_at IS NULL;
```

-- پایش و سختگیرانه کردن قوانین ایزوله‌سازی چندمستأجری در شرکای تجاری ۱.

```
ALTER TABLE business_partners
```

```
ALTER COLUMN tenant_id SET NOT NULL;
```

-- افزودن ستون‌های حیاتی اطلاعات تماس به جدول اشخاص حقیقی ۲.

```
ALTER TABLE persons
```

```
ADD COLUMN mobile VARCHAR(۲۰),
```

```
ADD COLUMN email VARCHAR(۲۵۵);
```

-- افزودن شناسه ملی/مالیاتی به اشخاص حقوقی ۳.

```
ALTER TABLE business_partner_organizations
```

```
ADD COLUMN tax_identifier VARCHAR(۵۰);
```

-- اصلاح جدول آرس‌ها (جایگزینی فیلدهای متنی استان/شهر با شناسه‌های ساختاریافته) ۴.

```
ALTER TABLE entity_addresses
DROP COLUMN IF EXISTS province_name,
DROP COLUMN IF EXISTS city_name,
ADD COLUMN province_id UUID, -- ارجاع منطقی به کاتالوگ تقسیمات کشوری
ADD COLUMN city_id UUID;
```

```
-- افزودن فلگ تایید هویت برای کانال‌های ارتباطی ۵. --
```

```
ALTER TABLE entity_contact_points
ADD COLUMN verified BOOLEAN NOT NULL DEFAULT FALSE;
```

```
-- اصلاح جداول کاتالوگ و دوره‌های عملیاتی سیستم ۶. --
```

```
ALTER TABLE fiscal_years ADD COLUMN currency_id UUID;
ALTER TABLE payment_terms ADD COLUMN installment_count INT NOT NULL DEFAULT ۱;
ALTER TABLE units_of_measure ADD COLUMN conversion_factor NUMERIC(۲۰,۴) NOT NULL
DEFAULT ۱,۰۰۰۰;
ALTER TABLE countries ADD COLUMN iso_numeric_code VARCHAR(۳);
```

```
-- ۷. Tenant سازمان‌ها و شرکت‌های زیرمجموعه Master Data companies --
```

```
-- =====
```

```
-- Layer ۵: ERP Core Master Data Layer
```

```
-- Organization Master Data
```

```
-- ۸. companies
```

```
-- =====
```

```
CREATE TABLE companies (
```

company_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

tenant_id UUID NOT NULL,

code VARCHAR(8) NOT NULL,

name VARCHAR(255) NOT NULL,

legal_name VARCHAR(255),

registration_number VARCHAR(100),

tax_identifier VARCHAR(100),

company_type SMALLINT NOT NULL DEFAULT 1,

-- 1: Legal Entity

-- 2: Business Unit

currency_id UUID,

status SMALLINT NOT NULL DEFAULT 1,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),

created_by UUID,

updated_at TIMESTAMPTZ,

updated_by UUID,

deleted_at TIMESTAMPTZ,

```
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_companies_tenant_code  
ON companies(tenant_id, code)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_companies_tenant  
ON companies(tenant_id)  
WHERE deleted_at IS NULL;
```

```
-- =====  
-- V. branches (Master Data شعبہ شرکتها)  
-- =====
```

```
CREATE TABLE branches (  
    branch_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  
    tenant_id UUID NOT NULL,  
  
    company_id UUID NOT NULL,
```

```
code VARCHAR(10) NOT NULL,  
name VARCHAR(20) NOT NULL,  
  
branch_type SMALLINT NOT NULL DEFAULT 1,  
  
manager_user_id UUID,  
  
status SMALLINT NOT NULL DEFAULT 1,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT 1  
);
```

```
CREATE UNIQUE INDEX uq_branches_company_code  
ON branches(company_id, code)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_branches_tenant
```

```
ON branches(tenant_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_branches_company
ON branches(company_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ۸. departments (Master Data واحدها و دپارتمان‌های سازمان)
-- =====
```

```
CREATE TABLE departments (
    department_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    company_id UUID NOT NULL,
    branch_id UUID,

    parent_department_id UUID,

    code VARCHAR(۵۰) NOT NULL,
    name VARCHAR(۲۰۰) NOT NULL,

    manager_employee_id UUID,
```

```
status SMALLINT NOT NULL DEFAULT \,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT \  
);
```

```
CREATE UNIQUE INDEX uq_departments_company_code  
ON departments(company_id, code)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_departments_tenant  
ON departments(tenant_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_departments_company  
ON departments(company_id)  
WHERE deleted_at IS NULL;
```



```
CREATE INDEX idx_departments_branch
ON departments(branch_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_departments_parent
ON departments(parent_department_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- ٩. cost_centers (Master Data مراکز هزینه)
-- =====
```

```
CREATE TABLE cost_centers (
    cost_center_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    company_id UUID NOT NULL,
    department_id UUID,

    parent_cost_center_id UUID,

    code VARCHAR(٥٠) NOT NULL,
    name VARCHAR(٢٠٠) NOT NULL,
    description VARCHAR(٥٠٠),
```

```

manager_employee_id UUID,

status SMALLINT NOT NULL DEFAULT \,

created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
created_by UUID,

updated_at TIMESTAMPTZ,
updated_by UUID,

deleted_at TIMESTAMPTZ,
deleted_by UUID,

row_version BIGINT NOT NULL DEFAULT \
);

-- Replace idx_cost_centers_department in 02_Master_Data_Table_Definitions.sql

CREATE INDEX idx_cost_centers_department
ON cost_centers(department_id)
WHERE deleted_at IS NULL; -- Typo Fixed (deleted at -> deleted_at)

CREATE INDEX idx_cost_centers_tenant
ON cost_centers(tenant_id)
WHERE deleted_at IS NULL;

```

```
CREATE INDEX idx_cost_centers_company
ON cost_centers(company_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_cost_centers_department
ON cost_centers(department_id)
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_cost_centers_parent
ON cost_centers(parent_cost_center_id)
WHERE deleted_at IS NULL;
```

```
-- =====
-- Ȳ. tenant_users (Tenant Membership)
-- =====
```

```
CREATE TABLE tenant_users (
    tenant_user_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    tenant_id UUID NOT NULL,

    user_id UUID NOT NULL,

    employee_id UUID,

    is_owner BOOLEAN NOT NULL DEFAULT FALSE,
```

```
status SMALLINT NOT NULL DEFAULT \,  
  
created_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
created_by UUID,  
  
updated_at TIMESTAMPTZ,  
updated_by UUID,  
  
deleted_at TIMESTAMPTZ,  
deleted_by UUID,  
  
row_version BIGINT NOT NULL DEFAULT \  
);
```

```
CREATE UNIQUE INDEX uq_tenant_users  
ON tenant_users(tenant_id, user_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_users_user  
ON tenant_users(user_id)  
WHERE deleted_at IS NULL;
```

```
CREATE INDEX idx_tenant_users_tenant  
ON tenant_users(tenant_id)  
WHERE deleted_at IS NULL;
```

